Exam Paper 01

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1 Instructions

- Time Allowed: 120 Minutes.
- Total Marks: 100.
- Section A: 20 Multiple Choice Questions (2 marks each).
- Section B: 10 Open Problems (6 marks each).
- Show all steps for Section B.

2 Section A: Multiple Choice

1. Which statement is logically equivalent to $P \Rightarrow Q$?
   (a) $Q \Rightarrow P$
   (b) $\neg P \vee Q$
   (c) $P \wedge \neg Q$
   (d) $\neg Q \Rightarrow P$
2. If $|A| = n$, what is $|\mathcal{P}(A)|$?
   (a) n^2
   (b) 2^n
   (c) $2n$
   (d) $n!$
3. An equivalence relation is:
   (a) Reflexive, Antisymmetric, Transitive
   (b) Reflexive, Symmetric, Transitive
   (c) Irreflexive, Symmetric, Transitive
   (d) Reflexive, Symmetric, Antitransitive
4. $f : \mathbb{Z} \to \mathbb{Z}, f(n) = 2n$ is:
   (a) Injective only
   (b) Surjective only
   (c) Bijective
   (d) Neither
5. $\gcd(2^{50} - 1, 2^{20} - 1)$ is:
   (a) $2^{20} - 1$
   (b) $2^{10} - 1$
   (c) $2^5 - 1$
   (d) 1
6. Compute $5^3 \pmod 7$.
   (a) 2

(b) 4
(c) 5
(d) 6

7. In induction, proving $P(1)$ and $P(k) \Rightarrow P(k+2)$ proves $P(n)$ for:

   (a) All n
   (b) Odd n
   (c) Even n
   (d) None

8. Arrangements of "BANANA":

   (a) 6!
   (b) 6!/(3!2!1!)
   (c) 6!/3!
   (d) 2^6

9. Coefficient of x^2y^3 in $(x+y)^5$:

   (a) 5
   (b) 10
   (c) 20
   (d) 1

10. $|3+4i|$:

    (a) 7
    (b) 5
    (c) 25
    (d) $\sqrt{7}$

11. i in exponential form:

    (a) $e^{i\pi}$
    (b) $e^{i\pi/2}$
    (c) $e^{-i\pi/2}$
    (d) $e^{i2\pi}$

12. If $1+i$ is a root of $P(x) \in \mathbb{R}[x]$, another root is:

    (a) $1-i$
    (b) $-1+i$
    (c) $-1-i$
    (d) i

13. Sequence $a_n = (-1)^n$:

    (a) Converges to 0
    (b) Converges to 1
    (c) Bounded but Divergent
    (d) Unbounded

14. $\lim_{n\to\infty} \frac{3n^2+5n}{2n^2-1}$:

    (a) $3/2$
    (b) 0
    (c) ∞
    (d) $5/2$

15. If $\mathbf{u} \cdot \mathbf{v} = 0$, vectors are:

    (a) Parallel
    (b) Orthogonal
    (c) Equal
    (d) Collinear

16. Which is a Field?

    (a) $\mathbb{Z}$
    (b) $\mathbb{Z}_6$
    (c) $\mathbb{Q}$
    (d) $\mathbb{R}^2$

17. Injection $A \to B$ and $B \to A$ implies:

    (a) $A = B$
    (b) $|A| < |B|$

(c) $|A| = |B|$
(d) $A \subseteq B$

18. 13 pigeons, 12 holes. One hole has:

    (a) Exactly 1
    (b) Exactly 2
    (c) At least 2
    (d) 0

19. Negation of $\forall x, \exists y, x < y$:

    (a) $\exists x, \forall y, x \geq y$
    (b) $\forall x, \exists y, x \geq y$
    (c) $\exists x, \exists y, x < y$
    (d) $\exists x, \forall y, x < y$

20. $x^2 + y^2 + z^2 = 9$:

    (a) Line
    (b) Plane
    (c) Sphere radius 3
    (d) Sphere radius 9

3 Section B: Open Problems

3.1 Problem B1

Prove by induction for $n \geq 1$: $\sum_{i=1}^{n}(2i-1) = n^2$.

3.2 Problem B2

Let $f : \mathbb{R} \setminus \{3\} \to \mathbb{R} \setminus \{2\}$, $f(x) = \frac{2x+1}{x-3}$.

1. Prove f is injective.
2. Prove f is surjective.
3. Find $f^{-1}(x)$.

3.3 Problem B3

Prove $xRy \iff 4 \mid (3x + y)$ is an equivalence relation on $\mathbb{Z}$.

3.4 Problem B4

Solve for smallest positive x:

$$\begin{cases} x \equiv 2 \pmod 3 \\ x \equiv 3 \pmod 5 \\ x \equiv 2 \pmod 7 \end{cases}$$

3.5 Problem B5

Committee of 5 from 8 Math, 6 CS profs.

1. Total ways?
2. Ways with at least 1 Math prof?

3.6 Problem B6

Solve $z^3 = -27i$.

3.7 Problem B7

$P(x) \in \mathbb{R}[x]$ has roots 1 and $2 + i$. Find $P(x)$.

3.8 Problem B8

Prove $\lim_{n\to\infty} \frac{2n+1}{n+3} = 2$ using $\epsilon - N$ definition.

3.9 Problem B9

Equation of plane through $(1, 0, 0), (0, 2, 0), (0, 0, 3)$.

3.10 Problem B10

Prove $(0, 1)$ is uncountable.

4 Answers

4.1 Section A

Q	A	Q	A	Q	A	Q	A
1	B	6	D	11	B	16	C
2	B	7	B	12	A	17	C
3	B	8	B	13	C	18	C
4	A	9	B	14	A	19	A
5	B	10	B	15	B	20	C

4.2 Section B

4.2.1 B1: Induction

- **Base Case** ($n = 1$): LHS $\sum_1^1 (2(1) - 1) = 1$, RHS $1^2 = 1$. (1 pt)
- **Hypothesis**: Assume $\sum_{i=1}^{k} (2i - 1) = k^2$. (1 pt)
- **Step**: $\sum_{i=1}^{k+1} (2i - 1) = k^2 + (2(k + 1) - 1) = k^2 + 2k + 1 = (k + 1)^2$. (3 pts)
- **Conclusion**: Holds for all n. (1 pt)

4.2.2 B2: Functions

- **Injective**: $f(x) = f(y) \Rightarrow \frac{2x+1}{x-3} = \frac{2y+1}{y-3} \Rightarrow x = y$. (2 pts)
- **Surjective**: Solve $y = \frac{2x+1}{x-3} \to x = \frac{3y+1}{y-2}$, valid for $y \neq 2$. (2 pts)
- **Inverse**: $f^{-1}(x) = \frac{3x+1}{x-2}$. (2 pts)

4.2.3 B3: Relations

- **Reflexive**: $4|(3x + x) \to 4|4x$. True. (1.5 pts)
- **Symmetric**: Assume $4|(3x + y)$. Then $3x + y = 4x - (x - y)$ $x - y$ divisible by 4 $3y + x$ divisible by 4. (2.5 pts)
- **Transitive**: If $4|3x + y$ and $4|3y + z$, add: $3x + 4y + z \to 3x + z$ divisible by 4. (2 pts)

4.2.4 B4: System of Congruences

- $x = 3k + 2$
- $3k + 2 \equiv 3 \pmod 5 \rightarrow 3k \equiv 1 \rightarrow k = 2 \pmod 5$
- $k = 5j + 2$
- $x = 15j + 8$
- $15j + 8 \equiv 2 \pmod 7 \rightarrow j \equiv 1 \pmod 7$
- $x = 105m + 23$
- **Smallest positive solution**: **23**. (6 pts)

4.2.5 B5: Combinatorics

1. Total ways: $\binom{14}{5} = 2002$. (2 pts)
2. Forbidden case: all CS $\rightarrow \binom{6}{5} = 6$. (2 pts)
3. Result: $2002 - 6 = 1996$. (2 pts)

4.2.6 B6: Complex Numbers

- $z^3 = 27e^{i3\pi/2}$
- Roots: $z_k = 3e^{i(3\pi/2+2\pi k)/3}$
  - $k = 0$: $3i$
  - $k = 1$: $3(-\frac{\sqrt{3}}{2} - \frac{1}{2}i)$
  - $k = 2$: $3(\frac{\sqrt{3}}{2} - \frac{1}{2}i)$

(2 pts each)

4.2.7 B7: Polynomials

1. Third root: $\overline{2+i} = 2 - i$. (2 pts)
2. Coefficients:
   - $(x - (2 + i))(x - (2 - i)) = x^2 - 4x + 5$
   - $P(x) = (x - 1)(x^2 - 4x + 5)$
   - $P(x) = x^3 - 5x^2 + 9x - 5$
   - $a = -5,\ b = 9,\ c = -5$. (4 pts)

4.2.8 B8: Epsilon–N Limit

- Manipulation: $\left|\frac{2n+1}{n+3} - 2\right| = \frac{5}{n+3}$.
- Given $\epsilon > 0$, choose $N > \frac{5}{\epsilon} - 3$.
- For $n > N$: $\frac{5}{n+3} < \epsilon$. (6 pts)

4.2.9 B9: Plane Equation

- $\vec{AB} = (-1, 2, 0)$, $\vec{AC} = (-1, 0, 3)$. (2 pts)
- Normal: $\vec{n} = (6, 3, 2)$. (2 pts)
- Plane: $6(x-1) + 3y + 2z = 0 \to 6x + 3y + 2z = 6$. (2 pts)

4.2.10 B10: Uncountability

- Assume list $x_1, x_2, \ldots$
- Construct y differing from x_n in the n-th decimal place.
- y not in the list $\to$ contradiction. (6 pts)